

GNU-Plot-Lab-Notebook

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Section 1

Question 1

Plot the following functions for $-10 \leq x \leq 10$ on the same graph using GNUPLOT. Set 1000 samples. Take the range of y -values from -4 to $+4$. Mark each curve with different line styles and line widths. Label x -axis and y -axis as 'x-values' and 'y-values' respectively. Set the title of the plot as 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \coth(x) - \frac{1}{x}, \quad f_2(x) = e^{-\sin(x)}, \quad f_3(x) = x|x|$$

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set term png
3 set ou "C:/Users/Suparnajit2005/Documents/GitHub/SEC-lab/GNU-plot/Lab-files/Output
  /1/Q_xxi_ou.png"
4
5 set grid
6 set xzeroaxis linestyle 1 lc "black"
7 set yzeroaxis linestyle 1 lc "black"
8 set xrange [-10:10]
9 set yrange [-4:4]
10 set xlabel "x values"
11 set ylabel "y values"
12 set samples 1000
13
14 f1(x)=(1/(tanh(x)))-(1/x)
15 f2(x)=exp(-sin(x))
16 f3(x)=x*abs(x)
17
18 plot f1(x) lw 2 lc "#5f0f40" dt 1 , f2(x) lw 2 lc "#fb8b24" dt 2, f3(x) lw 2 lc "
  #9a031e" dt 3
```

The output plot is given in the following page.

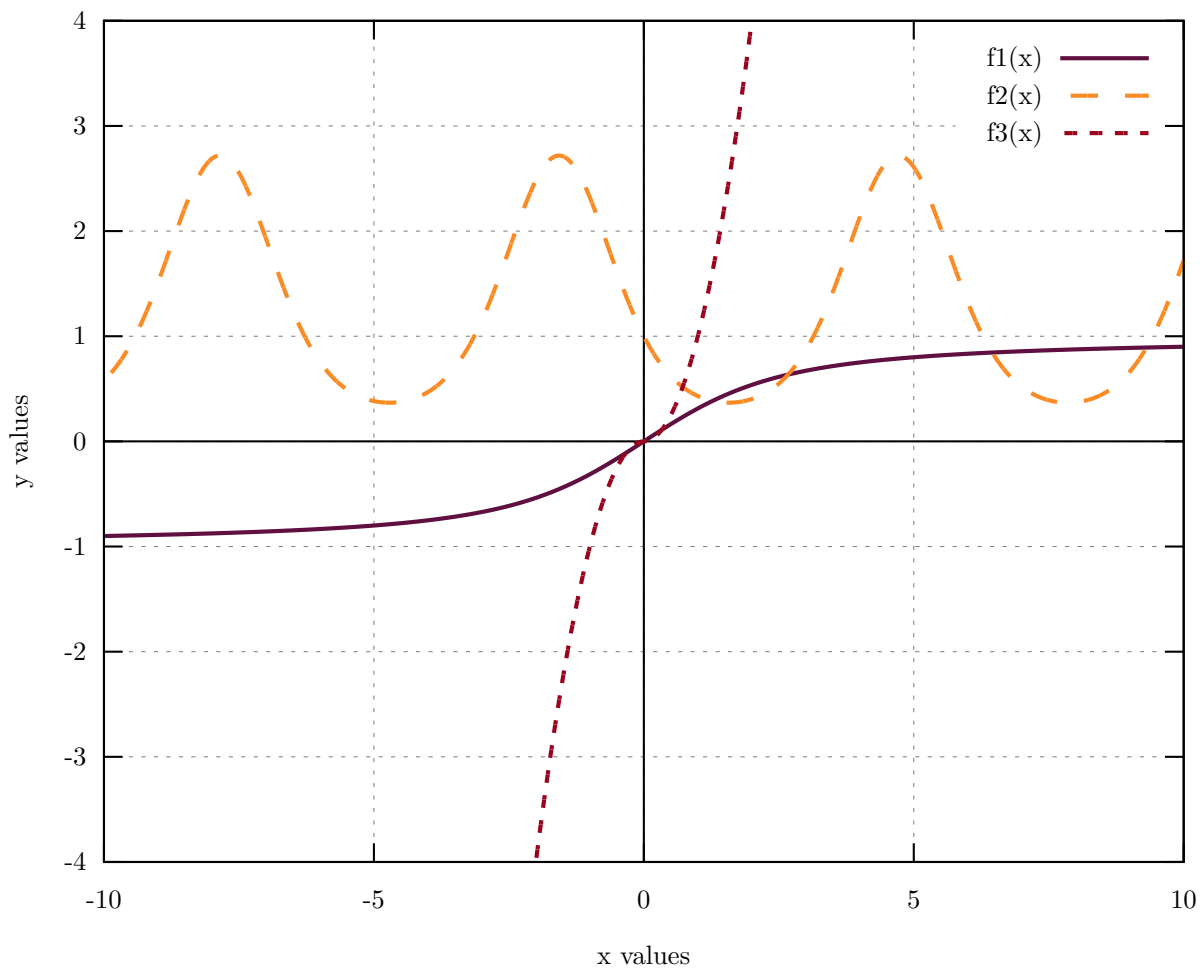


Figure 1: $f_1(x) = \coth(x) - \frac{1}{x}$, $f_2(x) = e^{-\sin(x)}$, $f_3(x) = x|x|$

Question 2

Plot the following functions for $0 \leq x \leq \pi$ on the same graph using GNUPLOT. Set 1000 samples. Mark each curve with different line style and line widths. Label x-axis and y-axis as 'x-values' and 'y-values' respectively. Set the title of the plot 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \sin^2(\sqrt{x}), \quad f_2(x) = e^{-x/2}, \quad f_3(x) = 12.34x^{5/3}$$

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set term svg background "white"
3 set ou "/home/suparnajit2005/GitHub/SEC-lab/GNU-plot/Lab-files/Output/1/Q_xx_ou.
  svg"
4
5 set title "Function plots"
6 set grid
7 set xzeroaxis linestyle 1 lc "black"
8 set yzeroaxis linestyle 1 lc "black"
9 set xrange [0:pi]
10 set yrange [0:8]
11 set xtics 0.5
12 set ytics 1
13 set xlabel "x-values"
14 set ylabel "y-values"
15
16 f1(x)=sin(sqrt(x))**2
17 f2(x)=exp(-x/2)
18 f3(x)=12.34*(x**(5/3))
19
20 set samples 1000
21 plot f1(x) lw 2 dt 1 lc "#780000" , f2(x) lw 2 dt 2 lc "#c1121f" , f3(x) lw 2 dt 3
  lc "#003049"
```

The output plot is given in the following page.

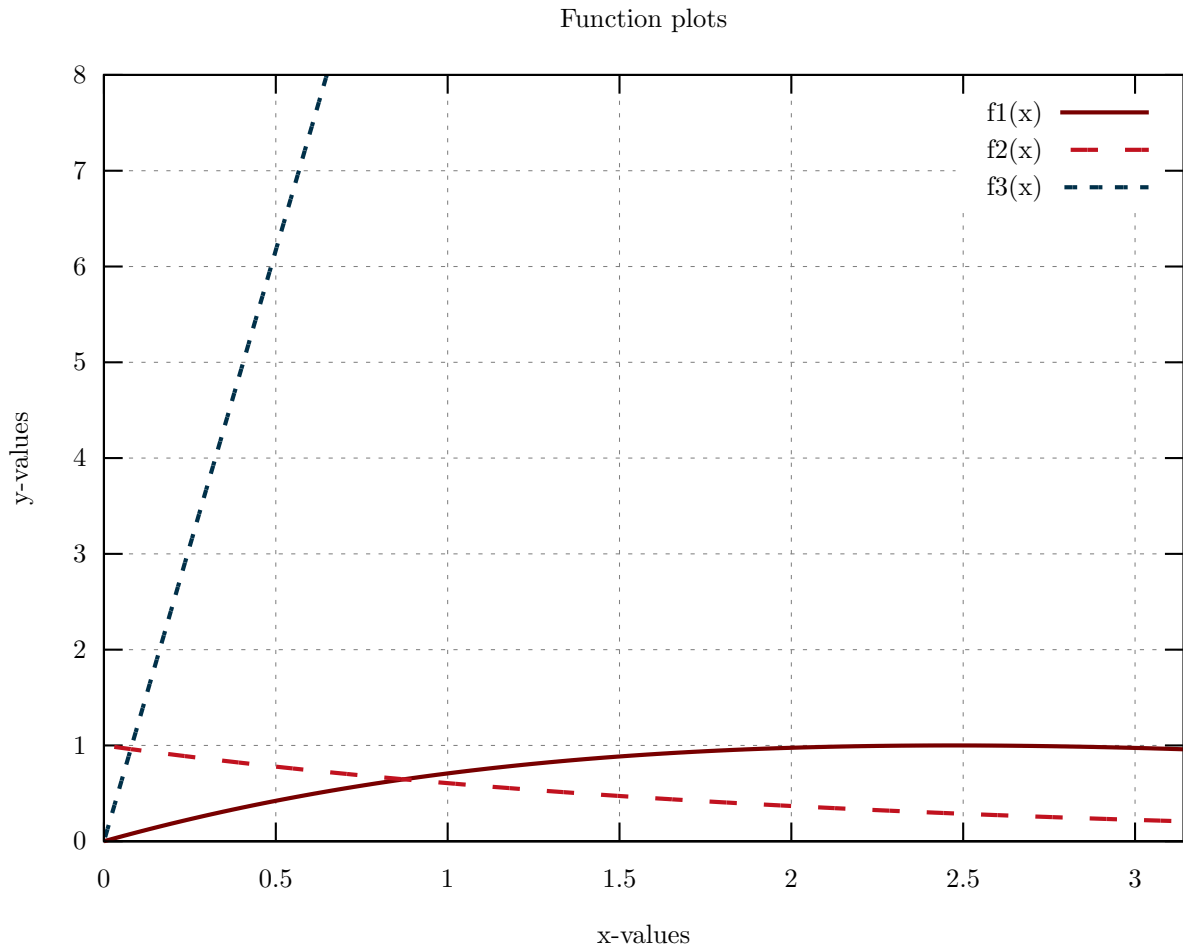


Figure 2: $f_1(x) = \sin^2(\sqrt{x})$, $f_2(x) = e^{-x/2}$, $f_3(x) = 12.34x^{5/3}$

Question 3

Define three functions in the given range:

$$\begin{aligned}f_1(x) &= \ln\left(\frac{1}{x}\right), & x > 0.1, \\f_2(x) &= 3x^2 \sin x, & -1 < x < 0.1, \\f_3(x) &= \cos 3x, & x < -1.\end{aligned}$$

Set the range of x values from -3 to $+3$ and the range of y values from -2.5 to $+2.5$. Show both the axes on the plot. Set y label “Piecewise continuous function”. Set ticks of the y axis at an interval of 0.5 . Plot the three functions and store the plot in a file and print it.

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set term svg background "white"
3 set ou "/home/suparnajit2005/GitHub/SEC-lab/GNU-plot/Lab-files/Output/1/Q_xiv_ou.
  svg"
4
5 set grid
6
7 set xzeroaxis linestyle 1 lc "black"
8 set yzeroaxis linestyle 1 lc "black"
9
10 set ytics 0.5
11
12 set xrange [-3:3]
13 set yrange [-2.5:2.5]
14
15 set ylabel "Piecewise continuous function"
16
17
18 f1(x)=(x>0.1)?(log(1/x)):NaN #Not as Number
19 f2(x)=(x<0.1 && x>-1)?(3*(x**2)*sin(x)):NaN
20 f3(x)=(x<-1)?(cos(3*x)):NaN
21
22 plot f1(x) lc "#9b2226" lw 2 , f2(x) lc "#bb3e03" lw 2 , f3(x) lc "#ee9b00" lw 2
```

The output plot is given in the figure below.

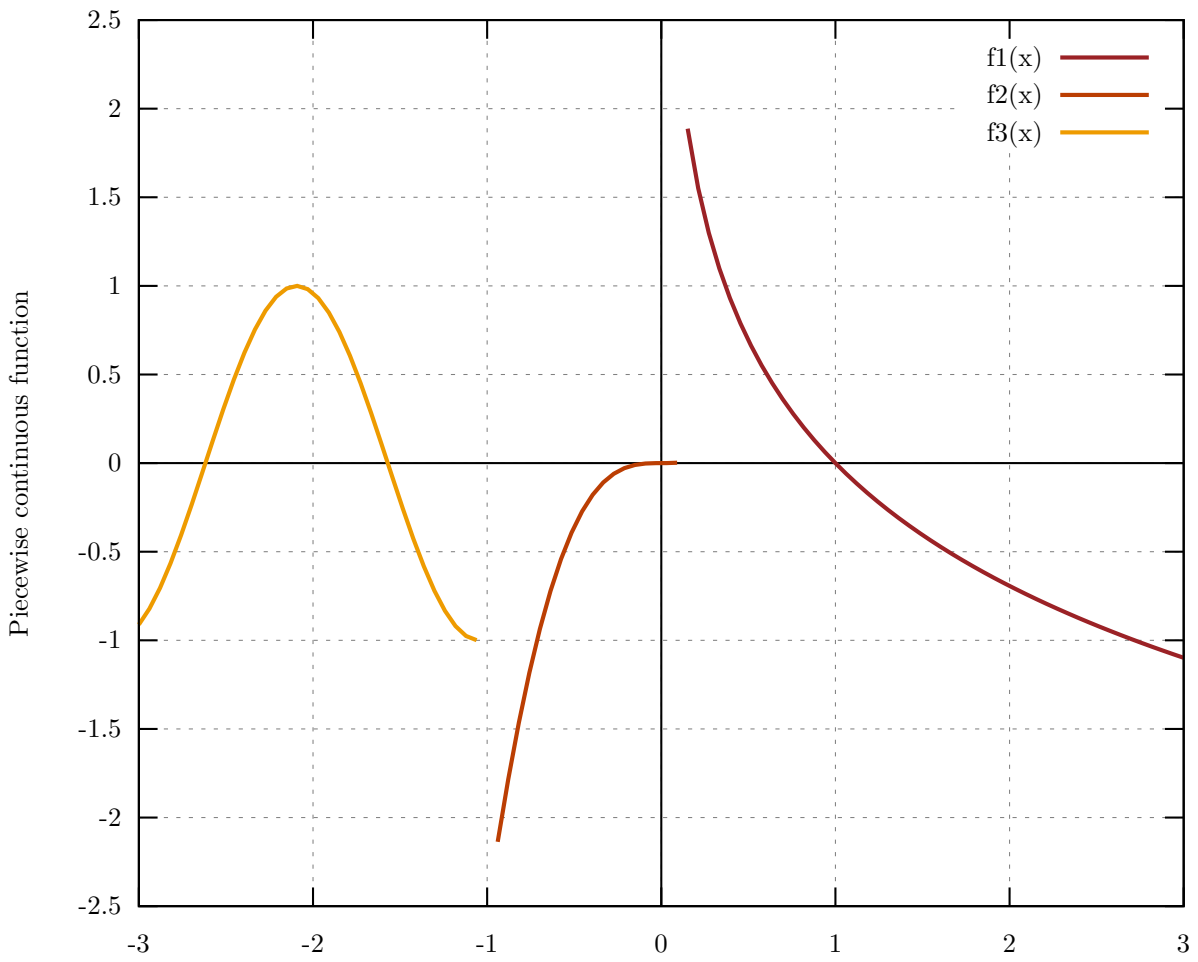


Figure 3: Plot of the piecewise continuous functions $f_1(x)$, $f_2(x)$, and $f_3(x)$

Question 4

Plot

$$2x \sin 2x, \quad \frac{1}{2}x \sinh\left(\frac{x}{2}\right)$$

on the same graph. The line width of the two curves to be different. Set the range of x values from $-\pi$ to $+\pi$ and the range of y values from -5 to 5 . Show both the axes on the plot. Ticks of x axis will be at interval of 0.5 . The label of x and y axis to be "x values" and "y values" respectively. Store the plot in a file and print it.

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set term svg background "white"
3 set ou "C:/Users/Suparnajit2005/Documents/GitHub/SEC-lab/GNU-plot/Lab-files/Output
  /1/Q_xvi_ou.svg"
4
5 set grid
6 set xzeroaxis linestyle 1 lc "black"
7 set yzeroaxis linestyle 1 lc "black"
8 set xrange [-pi:pi]
9 set yrange [-5:5]
10
11 f1(x)=2*sin(2*x)
12 f2(x)=(x/2)*sinh(x/2)
13
14 plot f1(x) lc "#9a031e" lw 2,f2(x) lc "#0f4c5c" lw 2
```

The output plot is given in the figure below.

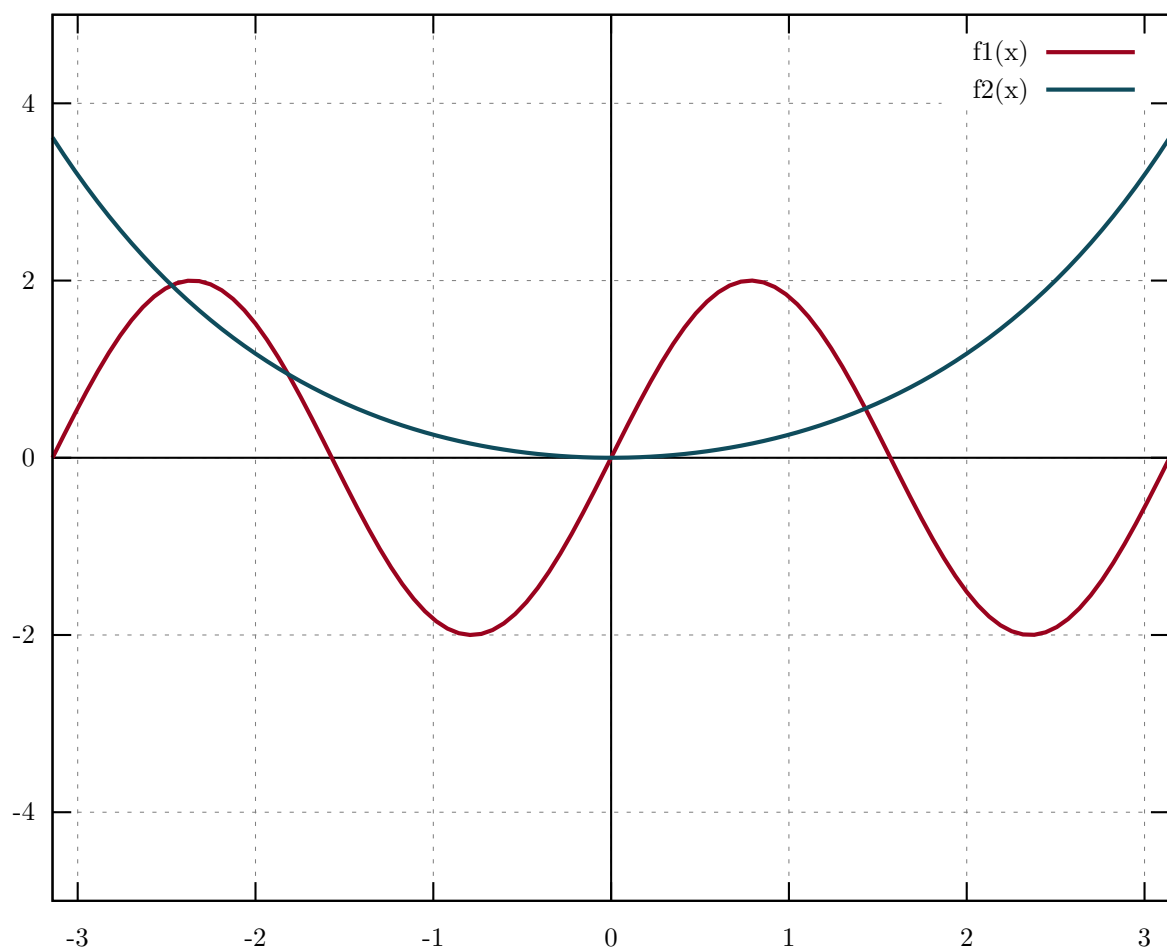


Figure 4: $2x \sin 2x$, $\frac{1}{2}x \sinh\left(\frac{x}{2}\right)$